

Analysis of 3rd-party Data Sharing and Data Tracking and of GDPR Compliance of Norwegian Web Sites

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ACIT4280 Privacy by Design

Group Assignment 1A

3 September 2026

Introduction

Group Assignment 1A has two parts. Part 1 maps third-party data sharing, cross-border contact and tracking on the 18 course-listed web services with Webbkoll. Part 2 (Group 1-A) uses the ICO lawful basis tool on five services chosen from that list. The 18 sites are Table 1. The five ICO sites are Table 1a. The Webbkoll numbers are Tables 2 and 3. Sector graphics are Figures 2 to 4.

The General Data Protection Regulation (GDPR) sets the rules for personal data in the EEA (General Data Protection Regulation, 2016). Processing is lawful only if at least one Article 6 basis applies. The bases used here are legal obligation, public task, contract, consent and legitimate interests. Cookies are also covered by ePrivacy rules (Cookie Directive, 2009; ePrivacy Directive, 2002). A third-party request is a call from the page to another host. That is not the same as a third-party cookie.

Table 1

Course Web Site List Analysed with Webbkoll

Sector (as on Canvas)	Services
Shopping	xxl.no, cdon.no, ikea.no, babyshop.no
Government	skatteetaten.no, barnevernet.no, altinn.no, lanekassen.no
News & media & entertainment	aftenposten.no, nrk.no, netflix.no, document.no, klassekampen.no, spotify.no, worldofwarcraft.com
Spare time / sports	dnt.no, fotball.no, skiforeningen.no

Note. All 18 names follow the Group Assignment 1A list. Webbkoll was run on each name as given, including redirects (for example ikea.no to ikea.com). document.no is on the course list with a content warning; it is included because it is on that list, not as a source of facts.

Table 1a

Five Services Chosen from Table 1 for the ICO Lawful Basis Tool

Service	Sector	Why this site
babyshop.no	Shopping	Retail checkout; high request count on the repeat scan
skatteetaten.no	Government	Public authority; tax and cookies as two purposes
netflix.no	News & media & entertainment	Paid subscription (contract)
document.no	News & media & entertainment	Widest country list; advertising analytics
fotball.no	Spare time / sports	Highest third-party request count in Table 2

Note. One site from shopping, one from government, two from news, media and entertainment, and one from spare time / sports. ICO results are Table 5.

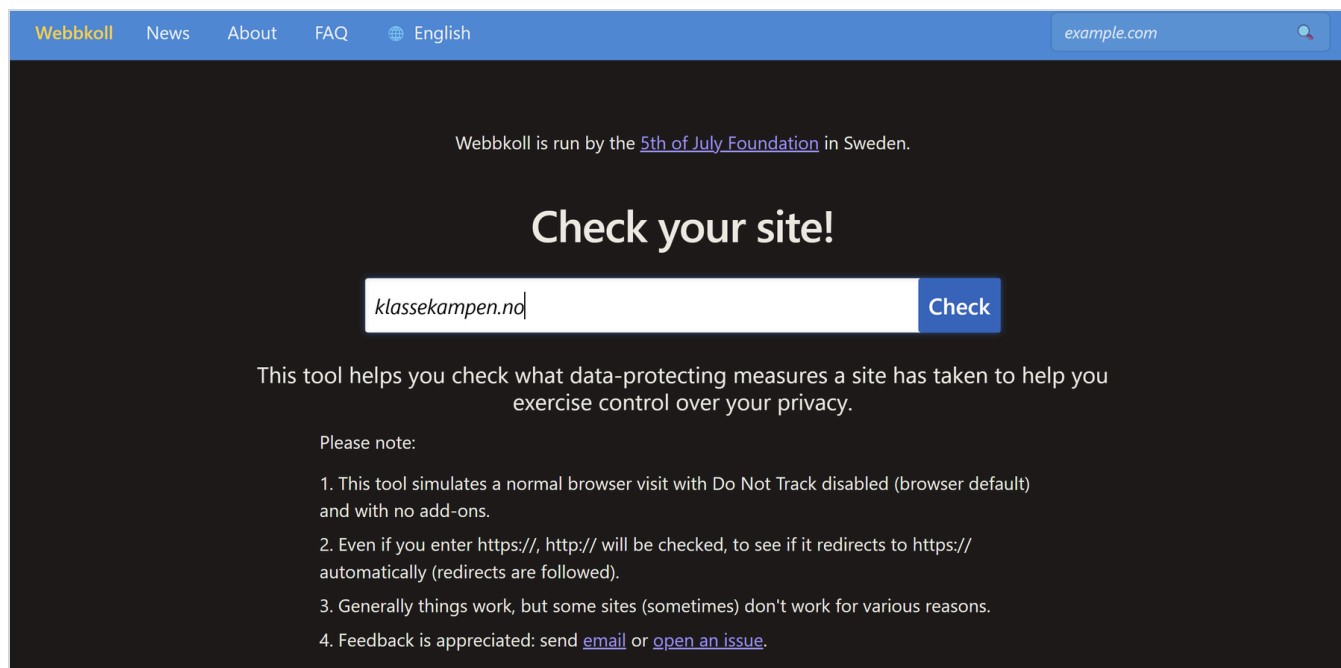
Method

Webbkoll is a public scanner (5th of July Foundation, n.d.-c). It reports HTTPS, cookies, third-party requests and the server IP. It does not read a privacy notice, click a banner, or decide Article 6. The English interface was used on every service in Table 1. Figure 1 shows the check page. Each result is one live Chromium visit with no add-ons, no Do Not Track and no consent click (5th of July Foundation, n.d.-a, n.d.-b). The scan time is recorded with the result.

Server location is the Country field in KeyCDN IP Location Finder (KeyCDN, n.d.). Where no country was returned, the cell is marked not available. Norway is excluded from the country count. Internal cookies are set by the site. External cookies are set by other parties. Third-party requests count other hosts contacted when the page loads. A site can contact many hosts without setting a third-party cookie. Table 2 is the first measurement. Table 3 is a repeat measurement. The two series are not averaged.

Figure 1

Webbkoll Start Page for klassekampen.no



Note. From *Webbkoll* [Screenshot], by 5th of July Foundation, n.d.-c (<https://webbkoll.dataskydd.net/en>).
klassekampen.no is fifth-highest on third-party requests in Table 2 and is therefore used as the method example.

Findings

Table 2 records the first measurement on 20 August 2026 for all 18 services in Table 1. Table 3 records a repeat measurement the same day, with KeyCDN countries for third-party hosts on 21 August 2026. A later visit can differ because Webbkoll stores a result for 24 hours.

Seventeen of 18 sites set zero third-party cookies. The exception is ikea.no (two cookies) after redirect to ikea.com. Third-party requests in Table 2 range from 1 (lanekassen.no) to 113 (fotball.no). Figure 2 ranks those counts. Figure 3 shows each site's share of the total. Figure 4 is the graphical overview per sector. News, media and entertainment contact the most hosts. Government sites contact the fewest. Spare time / sports is high because of fotball.no. Shopping lies between those extremes. babyshop.no rises to 101 requests on the repeat measurement. Webbkoll cannot prove that a request contains personal data. It can only show that a host in that country was contacted (General Data Protection Regulation, 2016, art. 44-47).

Table 2

Webbkoll Results for the 18 Services in Table 1 (First Measurement, 20 August 2026)

Name of service	Sector	Location of server	Internal cookies	External cookies	3rd-party requests	Countries excl. Norway	List of countries
xxl.no	Shopping	US	3	0	26	4	PL, IE, CA, SE

Name of service	Sector	Location of server	Internal cookies	External cookies	3rd-party requests	Countries excl. Norway	List of countries
cdon.no	Shopping	CA	4	0	33	3	CA, US, SE
ikea.no	Shopping	CA	8	2	15	2	CA, US
babyshop.no	Shopping	CA	1	0	45	2	SE, US
skatteetaten.no	Government	CA	5	0	14	5	IE, SE, FR, GB, NO, CA
barnevernet.no	Government	NO	2	0	25	4	SE, DE, CA, NO, GB
altinn.no	Government	CA	0	0	5	1	US
lanekassen.no	Government	NL	0	0	1	1	DE
aftenposten.no	News, media & entertainment	SE	5	0	48	5	SE, CA, US, DE, IE
nrk.no	News, media & entertainment	SE	10	0	7	1	SE
netflix.no	News, media & entertainment	GB	6	0	22	2	GB, CA
document.no	News, media & entertainment	DE	4	0	51	6	US, CA, SE, FR, DE, FI
klassekampen.no	News, media & entertainment	US	4	0	46	2	SE, US
spotify.no	News, media & entertainment	US	3	0	10	2	SE, US
worldofwarcraft.com	News, media & entertainment	IE	4	0	90	4	FR, SE, CA, US
dnt.no	Spare time / sports	GB	6	0	16	4	SE, CA, NL, FI
fotball.no	Spare time / sports	NL	8	0	113	5	SE, DK, CA, US, NL
skiforeningen.no	Spare time / sports	SE	1	0	14	3	NL, US, SE

Note. Columns follow the assignment table: name of service, location of server, internal cookies, external cookies, third-party requests, number of countries excluding Norway, and country list. Sector is added so Figure 4 can be read against Table 1. Lists may still show NO when it appeared on the first worksheet; the count column excludes it. Service names follow Table 1 even when Webbkoll followed a redirect.

Table 3

Repeat Webbkoll Measurement (20 August 2026) and KeyCDN Third-Party Countries (21 August 2026)

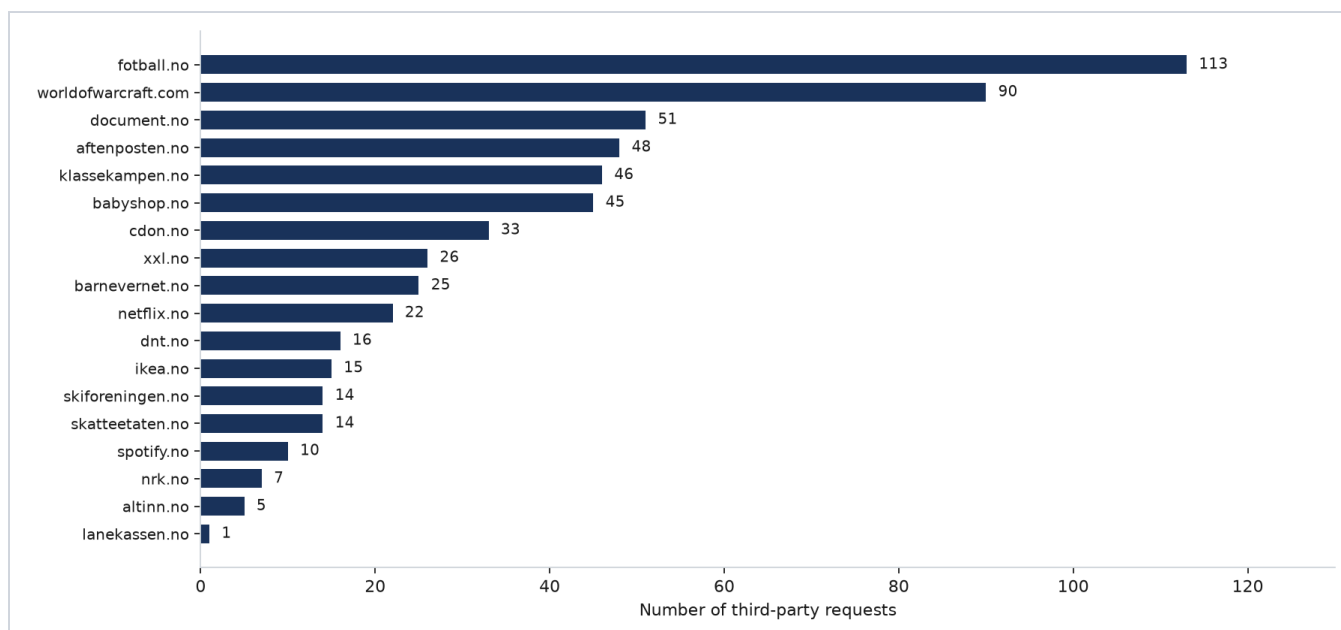
Name of service	Sector	Location of server	Internal cookies	External cookies	3rd-party requests	Countries excl. Norway	List of countries
xxl.no	Shopping	US	3	0	26	2	IE, US
cdon.no	Shopping	US	4	0	33	1	US

Name of service	Sector	Location of server	Internal cookies	External cookies	3rd-party requests	Countries excl. Norway	List of countries
ikea.no	Shopping	US	8	2	15	1	US
babyshop.no	Shopping	n/a	1	0	101	2	IE, US
skatteetaten.no	Government	US	5	0	14	4	FR, IE, SE, US
barnevernet.no	Government	NO	2	0	25	4	DE, IE, SE, US
altinn.no	Government	n/a	0	0	5	1	US
lanekassen.no	Government	NL	0	0	1	1	DE
aftenposten.no	News, media & entertainment	US	5	0	49	4	DE, IE, SE, US
nrk.no	News, media & entertainment	SE	11	0	7	1	SE
netflix.no	News, media & entertainment	US	6	0	22	1	US
document.no	News, media & entertainment	DE	4	0	51	5	FI, FR, IE, SE, US
klassekampen.no	News, media & entertainment	US	4	0	56	3	IE, SE, US
spotify.no	News, media & entertainment	US	3	0	10	2	SE, US
worldofwarcraft.com	News, media & entertainment	IE	4	0	92	3	IE, SE, US
dnt.no	Spare time / sports	n/a	6	0	16	5	FI, IE, NL, SE, US
fotball.no	Spare time / sports	NL	9	0	114	5	DK, IE, NL, SE, US
skiforeningen.no	Spare time / sports	US	1	0	14	3	IE, NL, US

Note. Table 2 is unchanged. babyshop.no is 101 versus 45; klassekampen.no is 56 versus 46. n/a means KeyCDN did not return a country (dnt.no, babyshop.no, altinn.no). Redirects were measured at the landing host.

Figure 2

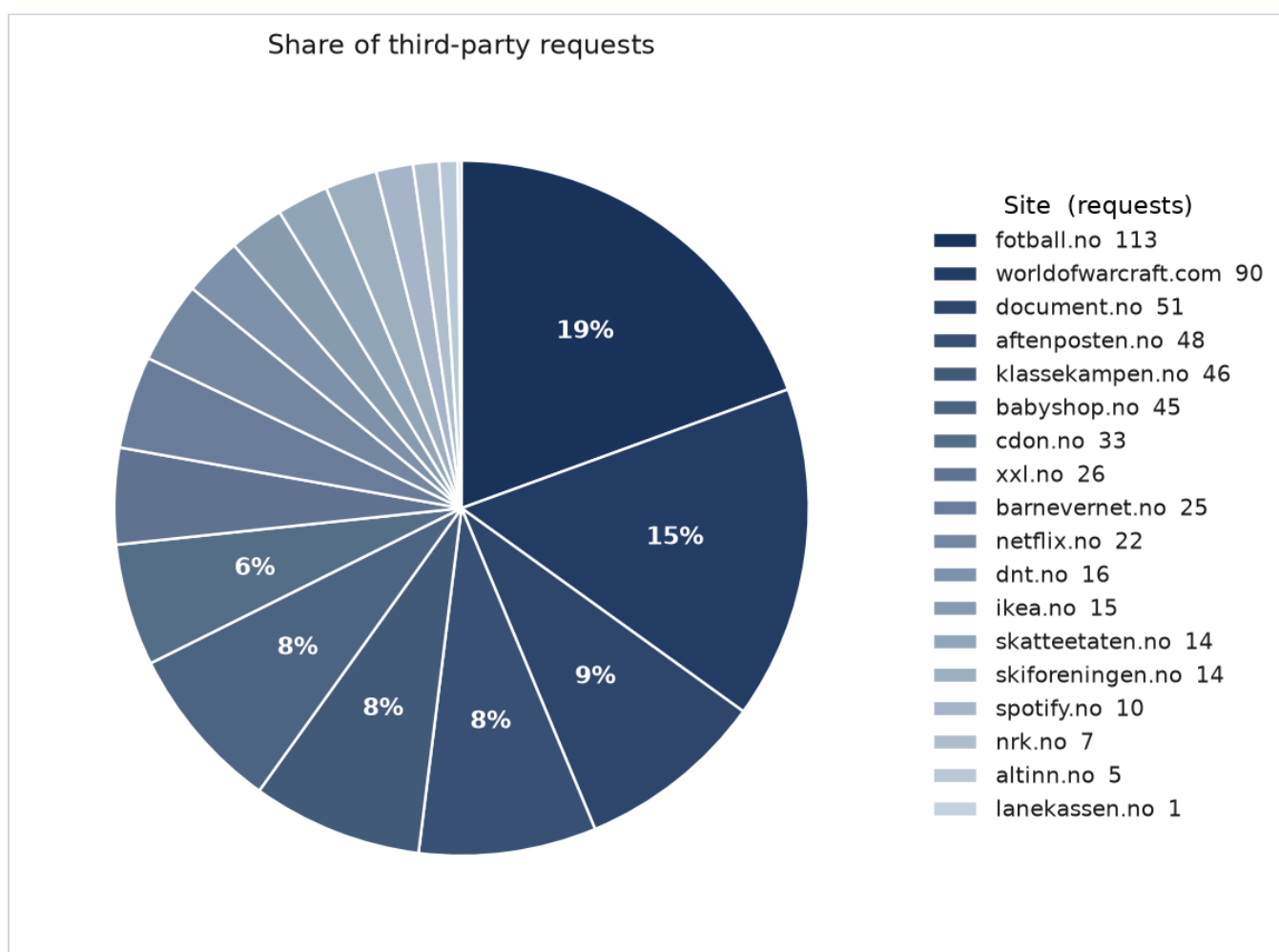
Third-Party Requests for 18 Sites (Table 2)



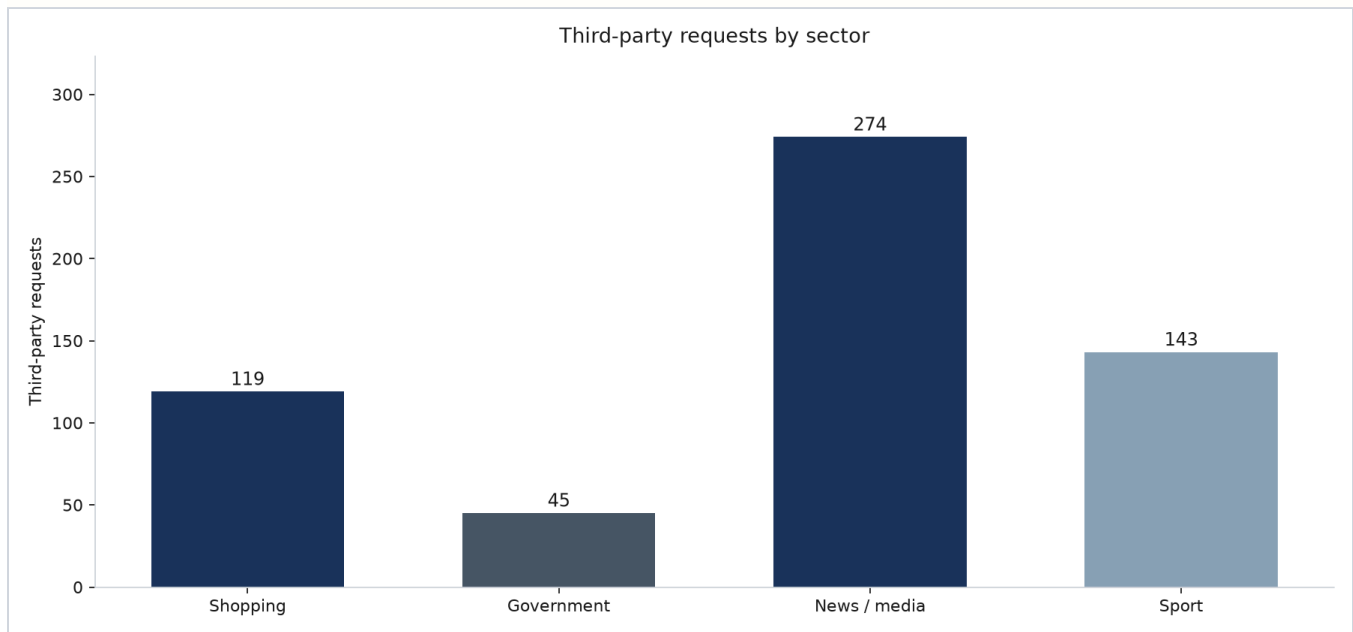
Note. Compiled by the authors from Webbkoll results. Ranked on the Table 2 request column. Highest five start at fotball.no (113). Lowest five start at lanekassen.no (1).

Figure 3

Share of Third-Party Requests (Table 2)



Note. Compiled by the authors from Webbkoll results. fotball.no and worldofwarcraft.com together account for about one third of all recorded requests.

Figure 4*Third-Party Requests by Sector (Table 2)*

Note. Compiled by the authors from Webbkoll results. Sector totals follow Table 2. News and media carry the most requests; government carries the fewest.

Sites of interest

The assignment asks for interesting findings such as the worst data sharer, the most globalised service, and the most privacy-friendly e-government site. Rank is third-party requests in Table 2 (Figure 2), not the external-cookie count. The worst data sharer on that measure is fotball.no (113 requests, still zero third-party cookies). The most globalised first scan is document.no (six countries excluding Norway). The most restrained e-government fronts are lanekassen.no (1 request) and altinn.no (5). Table 4 lists the highest five and the lowest five. A low cookie count can coexist with a high request count. The repeat measurement would place babyshop.no at 101 requests (Table 3). Rankings in this paper remain those of the first measurement.

Table 4*Highest Five and Lowest Five by Third-Party Request Volume (Table 2)*

Rank	Name of service	Sector	Req.	n countries
Highest 1	fotball.no	Spare time / sports	113	5
Highest 2	worldofwarcraft.com	News, media & entertainment	90	4
Highest 3	document.no	News, media & entertainment	51	6
Highest 4	aftenposten.no	News, media & entertainment	48	5
Highest 5	klassekampen.no	News, media & entertainment	46	2

Rank	Name of service	Sector	Req.	n countries
Lowest 5	skiforeningen.no	Spare time / sports	14	3
Lowest 4	spotify.no	News, media & entertainment	10	2
Lowest 3	nrk.no	News, media & entertainment	7	1
Lowest 2	altinn.no	Government	5	1
Lowest 1	lanekassen.no	Government	1	1

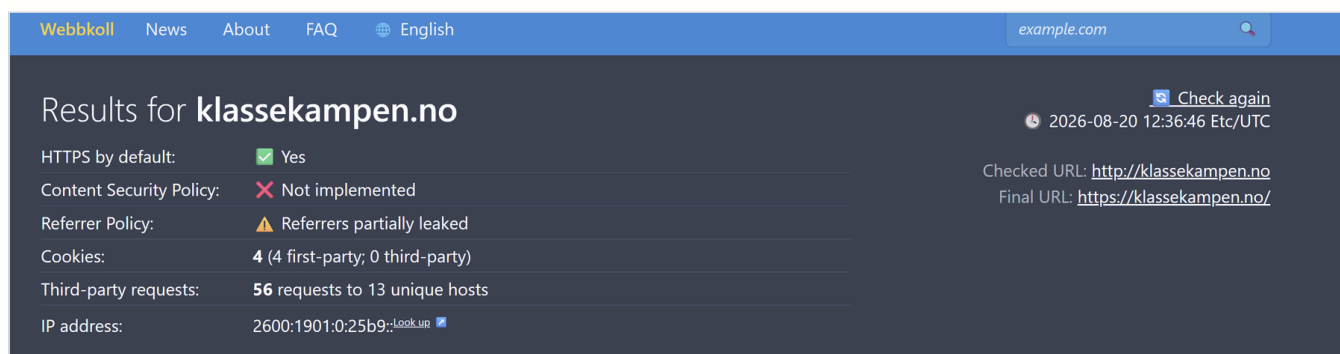
Note. Ranked on Table 2 requests. skatteetaten.no also has 14 requests; skiforeningen.no is listed because it has fewer countries (3 vs 5). document.no has the widest first country list (6). ikea.no remains the only third-party-cookie site (2 cookies).

Worked example: klassekampen.no

Figures 1, 5 and 6 show one full pass so the table rules can be checked. klassekampen.no is also Highest 5 in Table 4. Scan time: 12:36:46 UTC, 20 August 2026. Four first-party cookies and zero third-party cookies still came with 56 requests to 13 hosts, including Google advertising, Meta and Cookiebot. Missing HSTS, CSP and related headers concern integrity and confidentiality (General Data Protection Regulation, 2016, art. 5(1)(f), 25, 32). They do not decide the lawful basis. Privacy by design is both: how the page behaves, and why the controller may process the data. KeyCDN of the server IP is United States, Google (Figure 6).

Figure 5

Webbkoll Summary for klassekampen.no

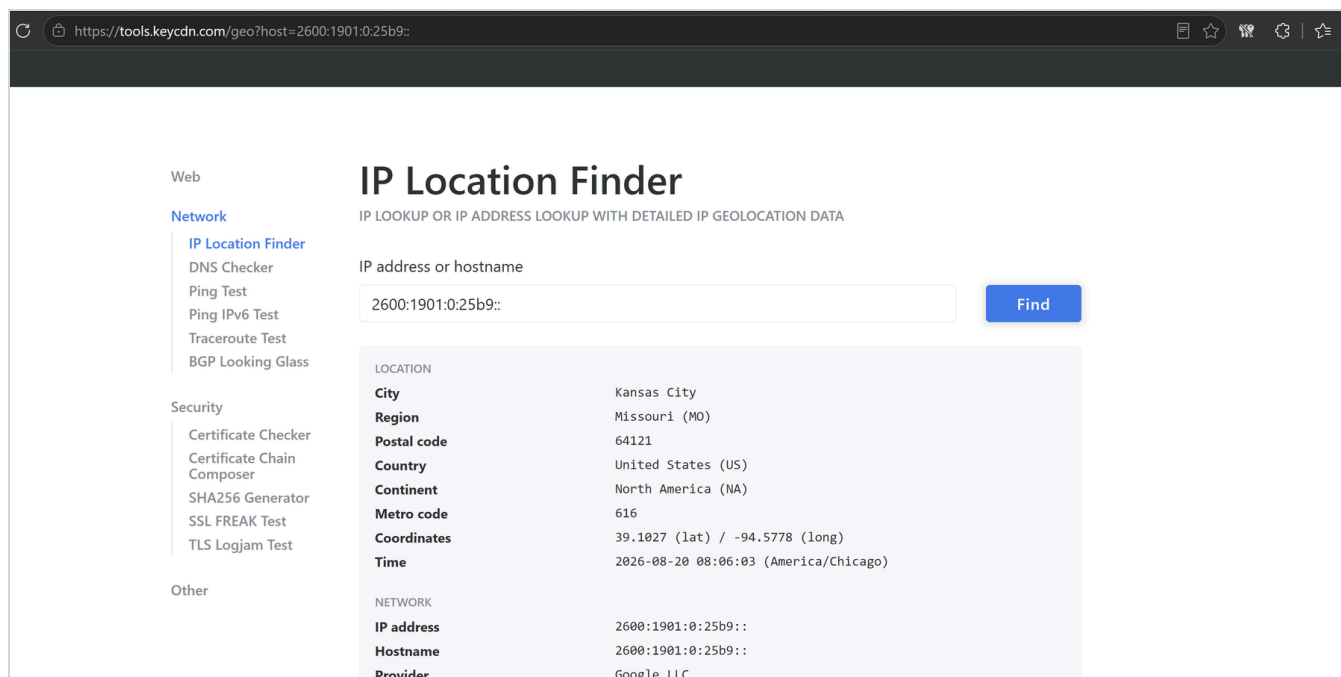


Webbkoll		News	About	FAQ	English	example.com
Results for klassekampen.no						Check again
						2026-08-20 12:36:46 Etc/UTC
HTTPS by default:	Yes					Checked URL: http://klassekampen.no
Content Security Policy:	Not implemented					Final URL: https://klassekampen.no/
Referrer Policy:	Referrers partially leaked					
Cookies:	4 (4 first-party; 0 third-party)					
Third-party requests:	56 requests to 13 unique hosts					
IP address:	2600:1901:0:25b9::lookur					

Note. From Webbkoll [Screenshot], by 5th of July Foundation, n.d.-c (<https://webbkoll.dataskydd.net/en>). Scan time 20 August 2026, 12:36:46 Etc/UTC. HTTPS by default: yes. CSP not implemented. Cookies: 4 first-party, 0 third-party. Third-party requests: 56 to 13 hosts.

Figure 6

KeyCDN Look Up of the klassekampen.no Server IP



Note. From *IP Location Finder* [Screenshot], by KeyCDN, n.d. (<https://tools.keycdn.com/geo>). Look up of 2600:1901:0:25b9::. Country: United States (US). Provider: Google LLC. The same tool on each unique third-party IP fills the country list.

Group 1-A: lawful basis for the five chosen services

From the 18 Webbkoll sites we chose five for the ICO tool (Table 1a): babyshop.no, skatteetaten.no, netflix.no, document.no and fotball.no. A service does not have one lawful basis for everything it does. The controller must name a purpose, then name a basis for that purpose (Information Commissioner's Office, n.d.). Each run states one concrete activity. Official ICO reports are dated 26 August 2026 (Information Commissioner's Office, 2026). Table 5 is the summary.

Article 6 is not a cookie click

Does a Yes/No banner decide whether the Tax Administration may process income data? No. Article 6 answers why personal data may be processed (General Data Protection Regulation, 2016, art. 6). Cookie rules answer whether information may be stored on a device (ePrivacy Directive, 2002). Mix the two in one ICO run, and the tool returns the wrong basis. For tax and the National Population Register, the notice says processing is required by law and is generally not optional (Norwegian Tax Administration, n.d.-a). The ICO report marks legal obligation and public task APPROPRIATE. Consent is NOT APPROPRIATE and likely invalid: the controller cannot offer a genuine free and ongoing choice for a statutory duty. Optional statistics cookies (Skyra, Matomo, Google Analytics, Siteimprove) are a

different purpose. They use a Yes/No banner. A dedicated ICO run marks consent INCONCLUSIVE: the banner offers a choice, but the request does not fully name controller, purpose and types. No basis is APPROPRIATE for that cookie purpose.

The other four runs keep one purpose. Netflix needs account and payment data to provide the paid service (Netflix, 2026). Contractual necessity means the processing is needed to perform the agreement, not merely useful. ICO marks contract APPROPRIATE and consent as likely invalid: there is no real, ongoing choice if the service cannot be delivered without the data. Norges Fotballforbund publishes name and club of active players aged 13+ as match history, with opt-out through the club (Norges Fotballforbund, n.d.). ICO marks legitimate interests APPROPRIATE. Membership in FIKS may rest on contract. Cookiebot may rest on ePrivacy consent (ePrivacy Directive, 2002). Those are other purposes. Babyshop's sales terms are a purchase contract (Babyshop Sthlm Holding AB, n.d.). ICO marks contract APPROPRIATE for checkout. The notice does not label Article 6(1)(b). Marketing and cookies stay separate.

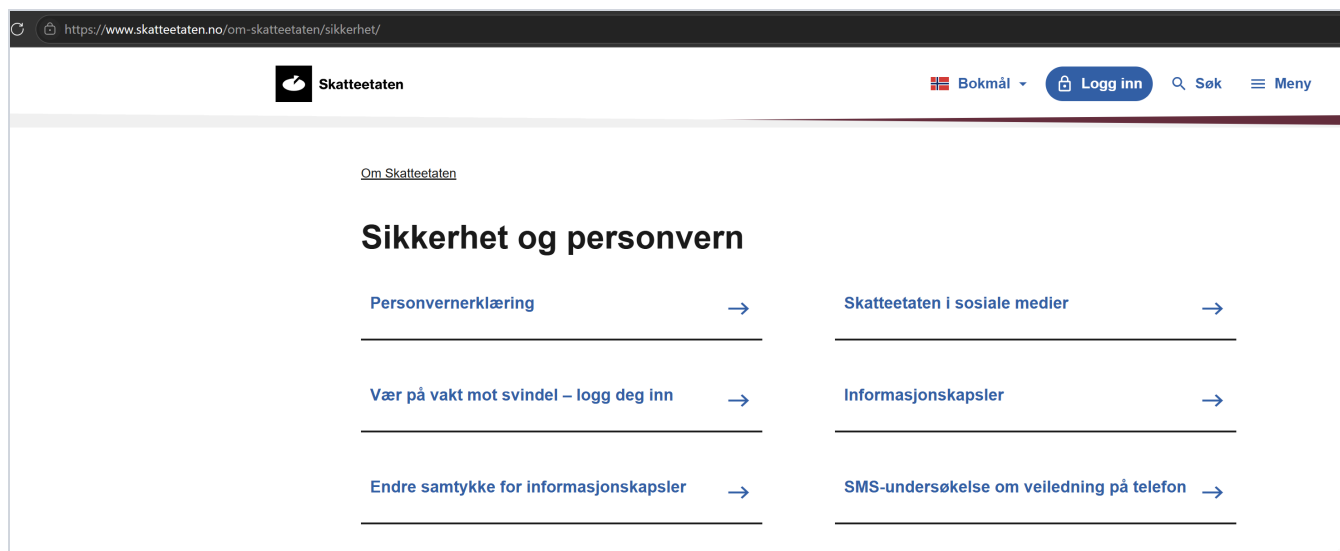
document.no is the case that fails the consent test. Google Signals is described as advertising analytics. Article 6 is not named (Document Media AS, n.d.). Collection is said to run only with a Google login and ad personalization. That looks like consent. The ICO report marks no basis APPROPRIATE. Consent is INCONCLUSIVE because a valid request must be clear, prominent and separate from other terms. The choice sits in Google settings. Document Pluss terms also bundle acceptance of the privacy notice. Contract and legitimate interests are not appropriate for this advertising purpose. A notice may aim at consent and still fail the request.

Optional cookies after No

Webbkoll does not click banners (5th of July Foundation, n.d.-b). skatteetaten.no/person/ was opened in a clean browser and No was selected on 24 August 2026. That tests whether optional tracking is blocked after rejection. It is not the lawful basis for tax administration. Figure 7 shows the Tax Administration hub that separates the privacy notice from cookie settings, which is the same split used in Table 5.

Figure 7

Security and Privacy Hub on skatteetaten.no



Note. From the Tax Administration security and privacy hub [Screenshot], by Norwegian Tax Administration, n.d.-b (<https://www.skatteetaten.no/en/about-the-norwegian-tax-administration/security-and-privacy/>). Personvernerklæring and Informasjonskapsler are listed as separate destinations. The two ICO rows for skatteetaten.no follow that split.

Table 5
ICO Lawful Basis Summary for Five Selected Services

Service	Processing purpose (one)	Basis claimed in notice	ICO result	Match?
skatteetaten.no	Tax and registry administration (identity, income, tax data)	Legal obligation / processing required by law; opt-out generally not possible	Legal obligation and public task APPROPRIATE; consent NOT APPROPRIATE (likely invalid)	Yes
skatteetaten.no	Optional web statistics cookies	Consent (Yes/No banner)	Consent INCONCLUSIVE; no basis APPROPRIATE	Partial (separate run)
netflix.no	Account and payment data to provide paid streaming	EEA/UK: contractual necessity to provide the service to members	Contract APPROPRIATE; other bases NOT APPROPRIATE	Yes
fotball.no	Publishing player name and club for match history (13+)	Public-interest match history with club opt-out (LI in practice)	Legitimate interests APPROPRIATE; other bases NOT APPROPRIATE	Yes for this purpose
document.no	Google Signals / advertising analytics from site activity	Article 6 not named; Google login plus ad personalization; opt-out links	Consent INCONCLUSIVE; no basis APPROPRIATE	Partial
babyshop.no	Customer, delivery and payment data to complete a purchase	Not labeled as Article 6(1)(b); purchase data processed to fulfil orders	Contract APPROPRIATE; other bases NOT APPROPRIATE	Yes for checkout

Note. One processing purpose per ICO row. Official ICO reports dated 26 August 2026 (Information Commissioner's Office, 2026). Marketing, cookies and membership administration stay separate purposes where the notice uses other bases.

Conclusion

The 18 course-listed sites are in Table 1. Webbkoll numbers for all of them are in Tables 2 and 3. Figure 4 is the sector overview. The worst data sharer on request volume is fotball.no. The widest country list is document.no. The most restrained e-government sites are lanekassen.no and altinn.no. The five ICO services (Table 1a) are babyshop.no, skatteetaten.no, netflix.no, document.no and fotball.no. Four of five core purposes match the notice (Table 5). Two consent tests fail: optional cookies on skatteetaten.no and Google Signals on document.no. A legal duty to assess tax is not a cookie click. One purpose requires one basis.

References

- Babyshop Sthlm Holding AB. (n.d.). *Integrity policy*. Babyshop. Retrieved August 26, 2026, from <https://www.babyshop.com/en-us/information/integrity-policy>
- Cookie Directive. (2009). Directive 2009/136/EC of the European Parliament and of the Council of 25 November 2009 amending Directive 2002/22/EC, Directive 2002/58/EC and Regulation (EC) No 2006/2004. EUR-Lex. <https://eur-lex.europa.eu/eli/dir/2009/136/oj>
- Document Media AS. (n.d.). *Personvernerklæring*. Document.no. Retrieved August 26, 2026, from <https://www.document.no/personvernerklaering/>
- ePrivacy Directive. (2002). Directive 2002/58/EC of the European Parliament and of the Council of 12 July 2002 concerning the processing of personal data and the protection of privacy in the electronic communications sector (Directive on privacy and electronic communications). EUR-Lex. <https://eur-lex.europa.eu/eli/dir/2002/58/oj>
- 5th of July Foundation. (n.d.-a). *About*. Webbkoll. Retrieved August 20, 2026, from <https://webbkoll.dataskydd.net/en/about>
- 5th of July Foundation. (n.d.-b). *FAQ*. Webbkoll. Retrieved August 20, 2026, from <https://webbkoll.dataskydd.net/en/faq>
- 5th of July Foundation. (n.d.-c). *Webbkoll* [Computer software]. Retrieved August 20, 2026, from <https://webbkoll.dataskydd.net/en>

General Data Protection Regulation. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation). EUR-Lex. <https://eur-lex.europa.eu/eli/reg/2016/679/oj>

Information Commissioner's Office. (2026). *Lawful basis interactive guidance tool*.

<https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/lawful-basis/lawful-basis-interactive-guidance-tool/>

Information Commissioner's Office. (n.d.). *A guide to lawful basis*. Retrieved August 26, 2026, from

<https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/lawful-basis/a-guide-to-lawful-basis/>

KeyCDN. (n.d.). *IP Location Finder* [Computer software]. Retrieved August 21, 2026, from

<https://tools.keycdn.com/geo>

Netflix. (2026, April 10). *Privacy statement*. <https://www.netflix.com/privacy>

Norges Fotballforbund. (n.d.). *Personvern*. Retrieved August 26, 2026, from

<https://www.fotball.no/personvern>

Norwegian Tax Administration. (n.d.-a). *Privacy policy*. Retrieved August 26, 2026, from

<https://www.skatteetaten.no/en/about-the-norwegian-tax-administration/security-and-privacy/privacy-policy/>

Norwegian Tax Administration. (n.d.-b). *Security and privacy*. Retrieved August 26, 2026, from

<https://www.skatteetaten.no/en/about-the-norwegian-tax-administration/security-and-privacy/>